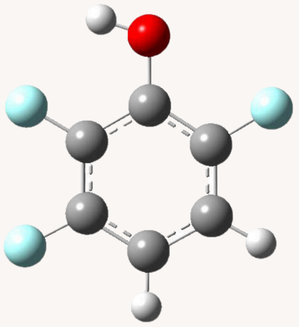
Microwave Spectrum and Analysis of 2,3,6-Trifluorophenol

By: Rohan Sarkar, Nadia Ahmed, Gordon Brown, Saravan Vishnubhotla,
South Carolina Governor's School for Science and Mathematics, Hartsville, SC 29550

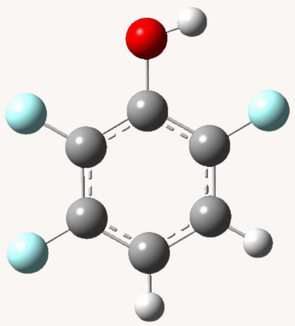
INTRODUCTION

- Spectroscopy is the study of the absorption and emission of light by matter. Atoms and molecules each have a unique spectrum. Spectroscopy is used to quantify information about molecules from these spectra.
- Microwave spectroscopy is the study between microwave radiation and molecules. It provides information about the rotational energy levels, which provides the structure of molecules.

2, 3, 6-Trifluorophenol



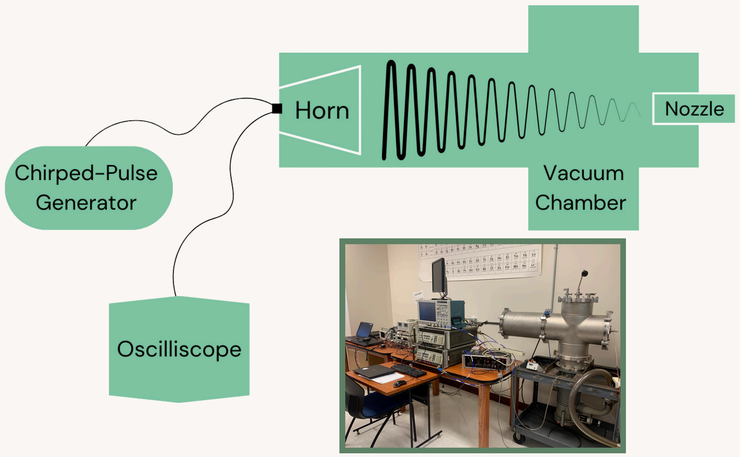
"cis"



"trans"

EQUIPMENT

Chirped-Pulse Fourier Transform Microwave Spectrometer

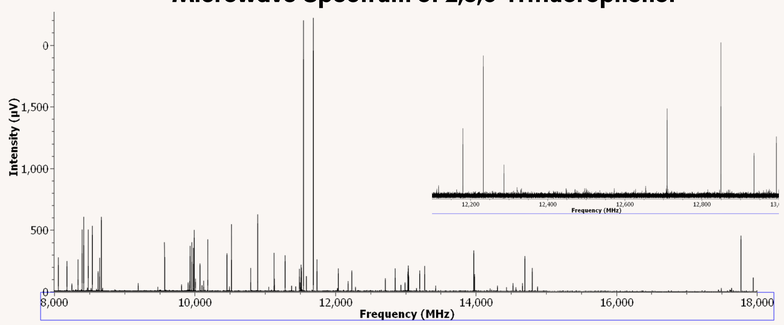


METHODOLOGY

- Placed 2, 3, 6-trifluorophenol in the nozzle of the microwave spectrometer at room temperature
- Measured the spectrum of the molecule using the microwave spectrometer
- Simulated molecule on Gaussian O3W software and calculated theoretical values of rotational and distortional constants (B3LYP/6-311+G(d,p))
- Used ASCP_L and SVIEW_L to assign quantum numbers to experimental transitions and obtain rotational and distortional constants

RESULTS

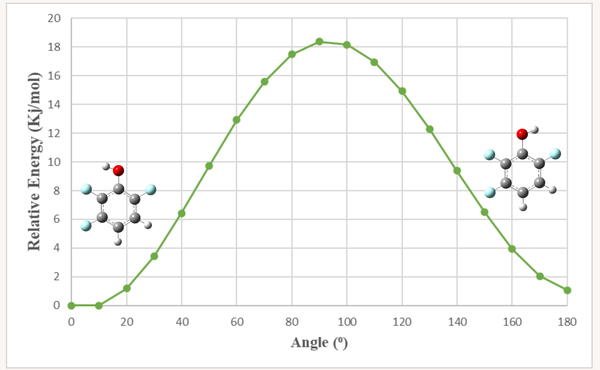
Microwave Spectrum of 2,3,6-Trifluorophenol



SPECTROSCOPIC CONSTANTS

	Experimental	*ab initio ("cis")*	*ab initio ("trans")*
A (MHz)	2122.9941 ± 0.0001	2096.1895	2120.8565
B (MHz)	1200.2975 ± 0.0001	1200.4002	1194.9000
C (MHz)	766.81992 ± 0.00007	763.29376	764.29361
DJ (kHz)	0.019 ± 0.001	0.020	0.020
DeIJK (kHz)	0.042 ± 0.005	0.047	0.046
DK (kHz)	0.085 ± 0.006	0.091	0.101
Dj (kHz)	0.0060 ± 0.0006	0.0069	0.0068
Dk (kHz)	0.031 ± 0.006	0.040	0.037
Relative Energy (Kj/mol)	N/A	0.0	1.1
Number of Lines	62 Lines	N/A	N/A
RMS (MHz)	0.001446	N/A	N/A

RESULTS CONT.



CONCLUSIONS

- Successfully measured microwave spectrum of 2, 3, 6-trifluorophenol
- Determined rotational and distortion constants for 2, 3, 6-trifluorophenol

FURTHER RESEARCH

- Remeasure the microwave spectrum from 15,000 to 17,500 MHz
- Research various chemical properties of 2, 3, 6-trifluorophenol
- Observe compound's interactions with various other molecules

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